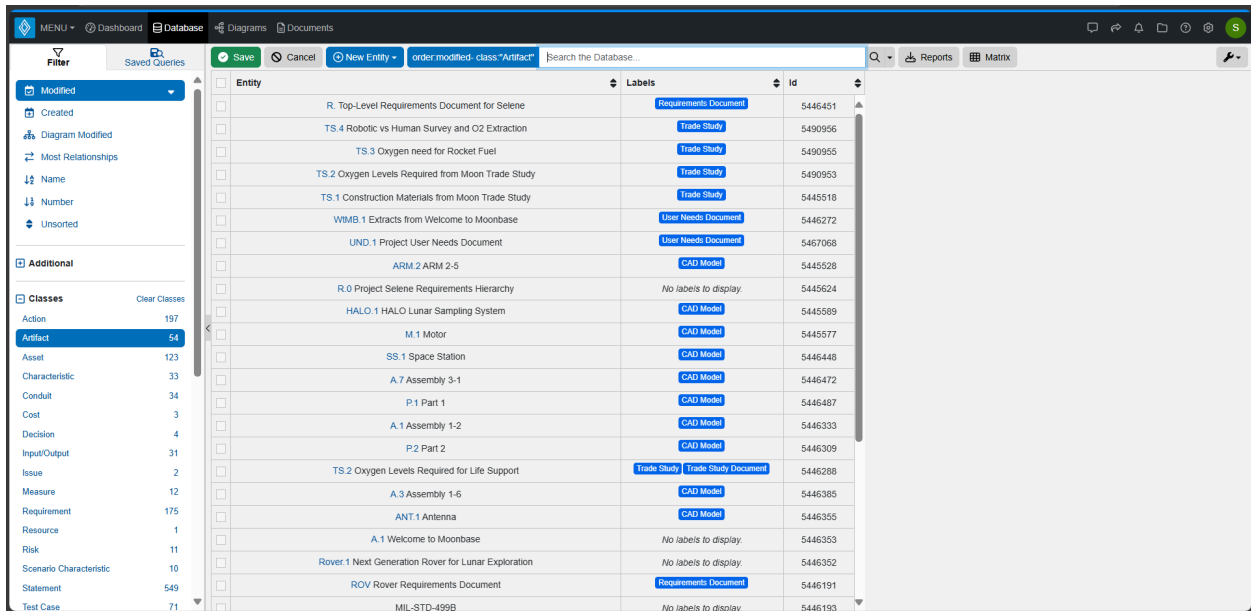


Lab-7

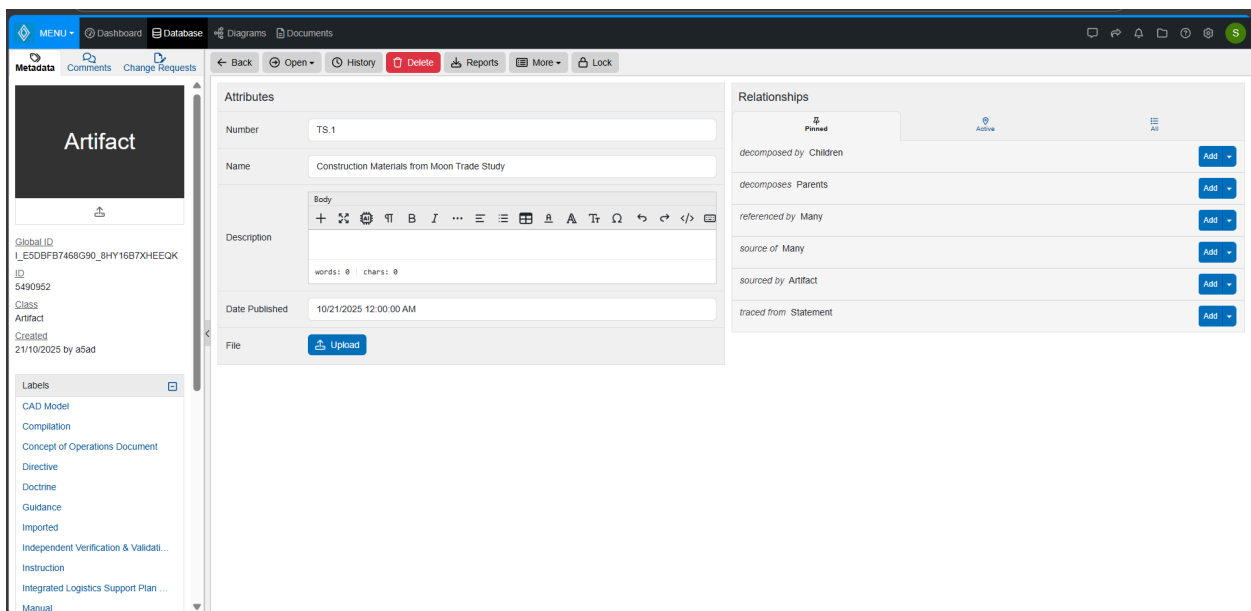
Syed M. Asad Ali (sa09475)

- Trade Study artifacts (TS.1–TS.4) created in the Database.



Entity	Labels	Id
R. Top-Level Requirements Document for Selene	Requirements Document	5446451
TS.4 Robotic vs Human Survey and O2 Extraction	Trade Study	5490956
TS.3 Oxygen need for Rocket Fuel	Trade Study	5490955
TS.2 Oxygen Levels Required from Moon Trade Study	Trade Study	5490953
TS.1 Construction Materials from Moon Trade Study	Trade Study	5446518
WMB.1 Extracts from Welcome to Moonbase	User Needs Document	5446272
UND.1 Project User Needs Document	User Needs Document	5467068
ARM.2 ARM 2-5	CAD Model	5445528
R.0 Project Selene Requirements Hierarchy	No labels to display.	5445624
HALO.1 HALO Lunar Sampling System	CAD Model	5445589
M.1 Motor	CAD Model	5445577
SS.1 Space Station	CAD Model	5446448
A.7 Assembly 3-1	CAD Model	5446472
P.1 Part 1	CAD Model	5446487
A.1 Assembly 1-2	CAD Model	5446333
P.2 Part 2	CAD Model	5446309
TS.2 Oxygen Levels Required for Life Support	Trade Study Trade Study Document	5446288
A.3 Assembly 1-6	CAD Model	5446385
ANT.1 Antenna	CAD Model	5446355
A.1 Welcome to Moonbase	No labels to display.	5446353
Rover.1 Next Generation Rover for Lunar Exploration	No labels to display.	5446352
ROV Rover Requirements Document	Requirements Document	5446191
MIL-STD-499B	No labels to display.	5446193

- Filled-in Trade Study entities with Name, Description, Date Published, and Label.



Artifact

Global ID: I_ESDBFB7468G90_8HY16B7XHEEQK
ID: 5490952
Class: Artifact
Created: 21/10/2025 by asad

Labels

- CAD Model
- Compilation
- Concept of Operations Document
- Directive
- Doctrine
- Guidance
- Imported
- Independent Verification & Validati...
- Instruction
- Integrated Logistics Support Plan ...
- Manual

Attributes

Number: TS.1

Name: Construction Materials from Moon Trade Study

Description:
Body:
words: 0 chars: 0

Date Published: 10/21/2025 12:00:00 AM

File: [Upload](#)

Relationships

- decomposed by: Children [Add](#)
- decomposes: Parents [Add](#)
- referenced by: Many [Add](#)
- source of: Many [Add](#)
- sourced by: Artifact [Add](#)
- traced from: Statement [Add](#)

Metastata

Comments

Change Request

Artifact

Global ID
L_6GAC47E1Z7GN5_B56RCDZZMRKM5

ID
5490953

Class
Artifact

Created
21/10/2025 by a5ad

Labels

Trade Study

CAD Model

Compilation

Concept of Operations Document

Directive

Doctrine

Guidance

Imported

Independent Verification & Validati...

Instruction

Diagrams

Documents

History

Delete

Reports

More

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Attributes

NumberTS 2

NameOxygen Levels Required from Moon Trade Study

Body

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Description

This trade study determines the range of oxygen needed to support life for the first phase of Moonbase operations. It sets the limit for meeting the verification requirements VR 1.3.1. It should also estimate levels needed for future phases of the program.

words: 41 | chars: 256

Date Published7/3/2024 12:48:52 PM

File

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Relationships

Pin

Active

All

decomposed by Children

Add

decomposes Parents

Add

referenced by Many

Add

source of Many

Add

sourced by Artifact

Add

traced from Statement

Add

http://cloud.lojistik.gov.tr/3772/diagrams

Metastata

Comments

Change Request

Artifact

Global ID
L_9VYTPTMXJCHK8_AR9C3QY8DN73

ID
5490955

Class
Artifact

Created
21/10/2025 by a5ad

Labels

Trade Study

CAD Model

Compilation

Concept of Operations Document

Directive

Doctrine

Guidance

Imported

Independent Verification & Validati...

Instruction

Integrated Logistics Support Plan ...

Diagrams

Documents

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Attributes

NumberTS 3

NameOxygen need for Rocket Fuel

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Description

This trade study determines the amount of oxygen needed for rocket fuel by the end of Phase1 of Moon base construction. It should also estimate needs for future phases of the program.

words: 32 | chars: 184

Date Published7/3/2024 9:40:53 AM

File

Upload

Relationships

Pin

Active

All

decomposed by Children

Add

decomposes Parents

Add

referenced by Many

Add

source of Many

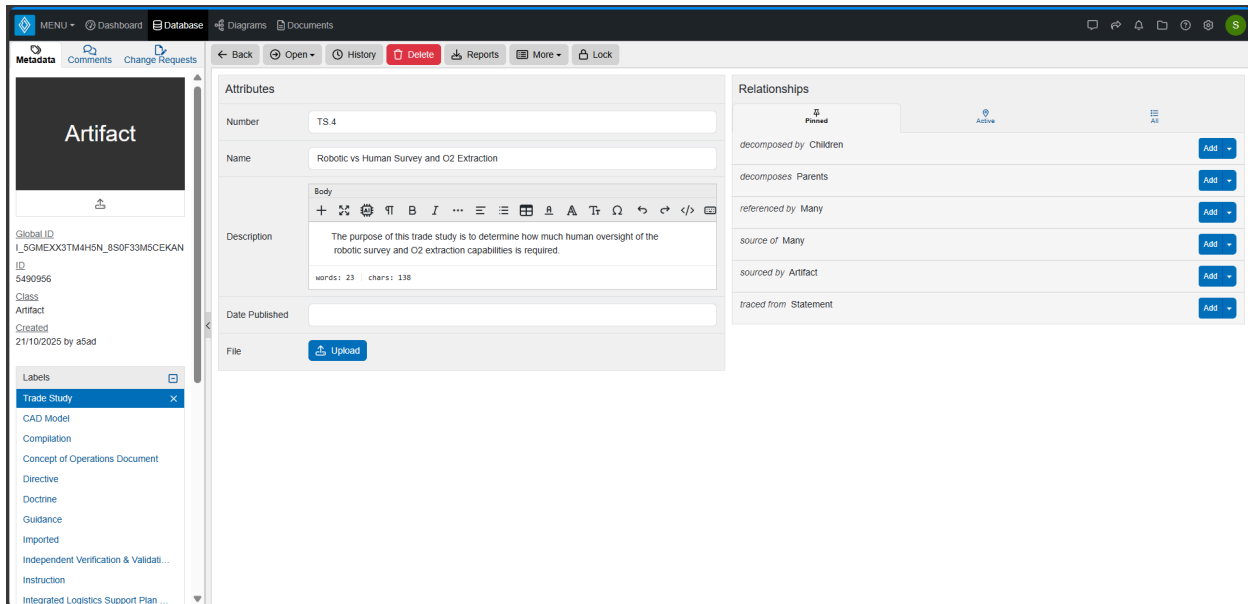
Add

sourced by Artifact

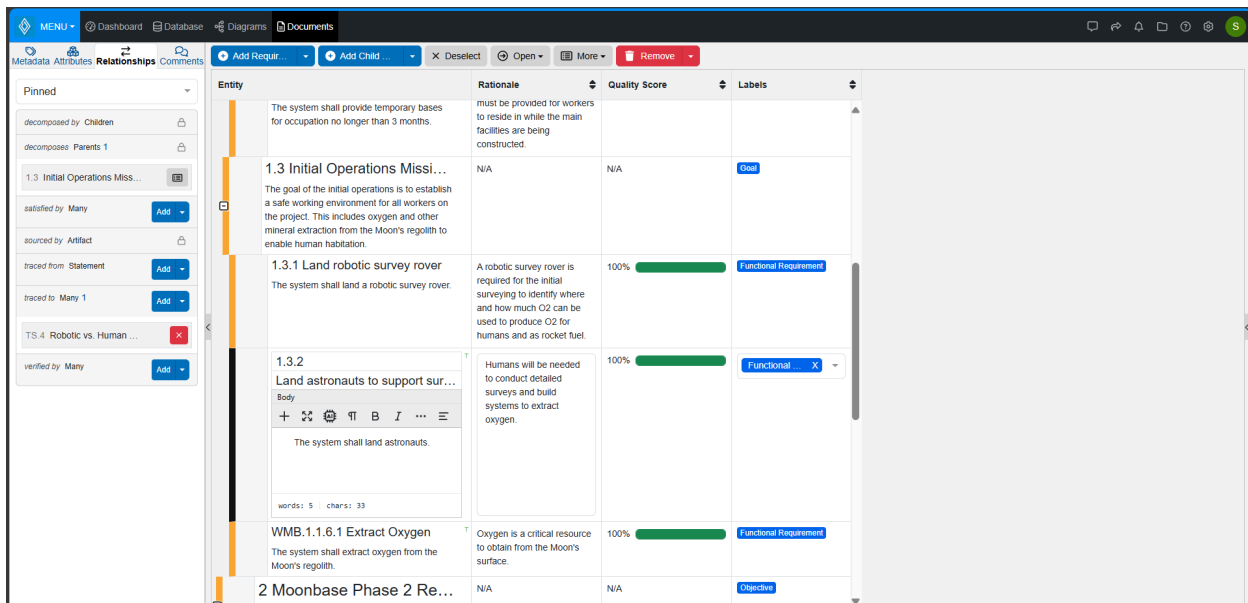
Add

traced from Statement

Add



- Screenshot showing connection of Trade Studies to corresponding Requirements.
- “Traced to” relationship between a Requirement and a Trade Study.



- Screenshot showing the Trade Studies inserted in the project document (Top-Level Requirements Document for Selene).

MENU

Dashboard

Database

Diagrams

Documents

Metadata

Attributes

Relationships

Comments

Add Requir...

Add Child...

Deselect

Open

More

Remove

Pinned

decomposed by Children

decomposed Parents 1

1.3 Initial Operations Miss...

satisfied by Many

sourced by Artifact

traced from Statement

traced to Many 1

TS.4 Robotic vs. Human ...

verified by Many

Entity	Rationale	Quality Score	Labels
The system shall provide temporary bases for occupation no longer than 3 months.	must be provided for workers to reside in while the main facilities are being constructed.		
1.3 Initial Operations Missi... The goal of the initial operations is to establish a safe working environment for all workers on the project. This includes oxygen and other mineral extraction from the Moon's regolith to enable human habitation.	N/A	N/A	Goal
1.3.1 Land robotic survey rover The system shall land a robotic survey rover.	A robotic survey rover is required for the initial surveying to identify where and how much O2 can be used to produce O2 for humans and as rocket fuel.	100%	Functional Requirements
1.3.2 Land astronauts to support sur... Body The system shall land astronauts. words: 5 chars: 33	Humans will be needed to conduct detailed surveys and build systems to extract oxygen.	100%	Functional ... X
WMB.1.1.6.1 Extract Oxygen The system shall extract oxygen from the Moon's regolith.	Oxygen is a critical resource to obtain from the Moon's surface.	100%	Functional Requirements
2 Moonbase Phase 2 Re...	N/A	N/A	Objective